Additional Problems for Ch 18 HRW – Temperature, Heat, Thermodynamics
Addisionele Probleme vir Hfst 18 HRW – Temperatuur, Warmte, Termodinamika

	9^{th} Edition	8^{th} Edition
Questions		
Problems	9, 12, 24, 36, 46, 50, 71	11, 10, 22, 30, 44, 48, 69

9	11	The new diameter is $D = 2.856\,\text{cm}$.	The new diameter is $D = 2.731\,\text{cm}$.
12	10	(a) $\Delta L = -2.49 \times 10^{-2}$ cm. $L' =$ 9.995 cm. (b) ΔT = 36 °C. Thus, T_x = 56°C.	(a) $\Delta L = -1.88 \times 10^{-2}$ cm. $L' =$ 9.996 cm. (b) ΔT = 48 °C. Thus, T_x = 68°C.
24	22	a) $c = 542\,\text{J/kg}\cdot\text{K}$. (b) The molar specific heat is given by $c_m = 27.1$ J/mol·K. (c) $N = \frac{m}{M} = \frac{30.0\times10^{-3}\text{ kg}}{50\times10^{-3}\text{ kg/mol}} = 0.600\,\text{mol}.$	a) $c = 523\,\text{J/kg}\cdot\text{K}$. (b) The molar specific heat is given by $c_m = 26.2$ J/mol·K. (c) $N = \frac{m}{M} = \frac{30.0\times10^{-3}\text{ kg}}{50\times10^{-3}\text{ kg/mol}} = 0.600\,\text{mol}.$
36	30	(a) $Q_w = 23.5\,\text{kcal}$. (b) $Q_b = 1.11\,\text{kcal}$. (c) $T_i = \frac{Q_w + Q_b}{c_c m_c} + T_f = 989°\text{C}$.	(a) $Q_w = 20.3\,\text{kcal}$. (b) $Q_b = 1.11\,\text{kcal}$. (c) $T_i = \frac{Q_w + Q_b}{c_c m_c} + T_f = 873°\text{C}$.
46	44	(a) W = –200 J. (b) Q = –80.0 cal = –335 J. (c) $\Delta E_{\text{int}} = Q - W$ = –293 J – (–200 J) = –135 J.	(a) W = –200 J. (b) Q = –70.0 cal = –293 J. (c) $\Delta E_{\text{int}} = Q - W$ = –293 J – (–200 J) = –93 J.
50	48	(a) Q_{ab} = +8.0 J. (b) Q_{bc} = (1.2 – 8.0 – 2.5) J = –9.0 J.	(a) Q_{ab} = +8.0 J. (b) Q_{bc} = (1.2 – 8.0 – 2.5) J = –9.3 J.
71	69	$c = \frac{\lvert Q\rvert}{m\,\lvert\Delta T\rvert} = 4.2\times10^{2}$ J/kg·K .	